

Ashitha Paramesha Gowda

Software Developer — Backend Developer — Cloud Engineer

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TECHNICAL SKILLS

- **Languages:** JavaScript, TypeScript, SQL, Java
- **Frameworks:** Node.js, NestJS, Express
- **Databases:** MongoDB, PostgreSQL
- **DevOps:** Docker, Kubernetes, Jenkins, GitHub Actions, CI/CD
- **Tools:** Git, Postman, REST API, Jest, Mocha, Chai, TDD, Jira, Confluence
- **Cloud:** AWS, Google Cloud Platform (GCP)
- **Blockchain:** Solidity, Web3.js, Hardhat, Ethers.js, Ganache, Truffle, IPFS, Smart Contracts

EXPERIENCE

Graduate Research Assistant (BSPH, JHU – Baltimore, MD)

Mar 2025 – Present

- Led technical requirement gathering with public health teams to define data extraction needs from Medicaid waiver PDFs across 15+ U.S. states.
- Developed a rule-based parser using Python, PyMuPDF, and regex to extract structured data from varied document layouts, reducing manual processing effort by 80%.
- Prototyped a bounding-box-based detection module using Google Document AI to improve extraction accuracy on complex, layout-heavy PDFs.

Software Developer (JKTech, Bengaluru)

May 2023 – July 2024

- Engineered folder-level file linking across S3, Box, and OneDrive using recursive indexing and pre-signed URLs, reducing manual file mapping by 30%.
- Designed and integrated OAuth-based Google Drive workflows with minimal scopes, boosting client onboarding satisfaction by 60%.
- Diagnosed and resolved a critical blockchain sync bug where the system misreported sync readiness; added validation checks and logged two new blockchain failure events to prevent data loss and improve user transparency.
- Authored test suites for event-driven flows with mock data and created fallback QA plans, increasing test coverage by 40% and reducing verification cycles by 25%.
- Resolved 80% of backend support tickets by tracing failures using Kibana logs and replaying event chains in EventStore, reducing average triage time by 2x.

Software Developer I (Juego Studios, Bengaluru)

Jan 2022 – Apr 2023

- Contributed to 14+ commercial projects including card games, betting platforms, and educational multiplayer apps; responsibilities included feature dev, bug fixes, and debugging legacy codebases.
- Designed and built a plug-and-play communication module for NJS2 framework supporting 1:1 and group messaging with voice via WebRTC; shipped with built-in room management and player moderation controls.
- Developed a secure device tracker app with LokiJS and SSO integration; added CLI flags `--local` and `--admin` to bypass SSO for internal testing, reducing admin request bottlenecks by 70%.
- Discovered and responsibly disclosed a server-side security bypass during internal auth debugging, enabling critical admin-level access fixes by the core infrastructure team.

Software Developer Intern (Juego Studios, Bengaluru)

Aug 2021 – Dec 2021

- Built a real-time multiplayer Snake & Ladder game from scratch using Node.js, Socket.io, and Argon; supported six-player rooms with latency-aware reconnection logic.
- Developed reusable backend handlers for client disconnects and rejoin flows, reducing session drop errors by 50%.
- Implemented dynamic frontend in Argon UI to manage real-time player turns, dice roll animations, and synchronized room state via Socket.io channels.
- Collaborated with senior developers to implement modular session architecture and track performance bottlenecks using test environments.

Founder, Full Stack Developer (CNotes.in, Bengaluru)

Jan 2021 – Present

- Built and deployed a cable TV CRM platform using PHP, MySQL, HTML/CSS, and Textlocal API, serving over 30,000 subscribers across 50+ operators.
- Developed billing, SMS receipts, and payment tracking workflows tailored for last-mile collection needs; improved on-time payments by 35%.
- Released a PWA version to support mobile operators on low-end devices; drove platform access up by 40% within the first quarter.
- Collected operator feedback through direct calls and field visits, integrating high-priority UX updates that led to a 90%+ customer retention rate.

EDUCATION

Johns Hopkins University

Master's in Computer Science

Vidyavardhaka College of Engineering

Bachelor of Engineering in Information Science

Baltimore, MD

Aug 2023 – May 2025

Karnataka, India

Aug 2017 – May 2021

PROJECTS

Patent Verification System (Johns Hopkins – Baltimore, MD)

Dec 2024 – May 2025

- Led a 3-member team as Full Stack Blockchain Architect to develop a secure patent verification platform.
- Designed and deployed smart contracts using Solidity and Hardhat on Ethereum Sepolia testnet.
- Integrated MetaMask authentication and wallet-based transaction signing for patent validation.
- Built the frontend using Next.js and integrated seamless Web3 connectivity.
- Implemented OAuth 2.0 for secure access and permission control.
- Stored IPFS hashes of patent files for decentralized verification and tamper resistance.
- Connected with OpenSea APIs for demonstration of patent ownership.
- Expected to reduce manual semantic review and document verification workload by 80%.

Clue Crafters Multiplayer Game (Johns Hopkins – Baltimore, MD)

Jan 2023 – May 2023

- Led a 4-member team and designed architecture for a turn-based detective board game supporting 6 players.
- Implemented NestJS and Express.js backend with WebSockets (Socket.io) for real-time game communication.
- Programmed cyclic turn logic, player movement based on virtual dice rolls, and accusation-based win conditions.
- Built React frontend with safeguards against invalid clicks and UI-state corruption.
- Developed and optimized reusable asset pipeline for character cards, weapons, and room tiles.
- Orchestrated session and room state management with role-based access to actions.
- Debugged and resolved high-impact gameplay logic bugs, especially around race conditions and session sync.
- Created internal documentation and diagrams for reproducibility and future scalability.

Personal Portfolio Website

2024

- Developed using Next.js and MUI to serve as a centralized showcase of technical projects and resume.
- Implemented dynamic routing with in-page rendering for SEO optimization and faster load times.
- Features project-specific navigation, including in-browser detail pages with back/forward memory.
- Highlights custom work like *XOXO AI Game*, an emotionally expressive variant of Tic-Tac-Toe powered by simple AI logic and behavioral variability.

Dear – Personalized Voice Assistant

Ongoing

- Designing an LLM-powered assistant to simulate emotionally resonant phone calls for individuals with Alzheimer's disease.
- Engineered a call flow system using scheduled triggers and voice cloning to provide comfort through familiar voices.
- Integrated speech-to-text and text-to-speech modules for bidirectional audio communication, with a focus on latency reduction and clarity.
- Fine-tuned prompts and memory systems in the LLM to mimic realistic, patient-specific conversations.
- Built using Python, AssemblyAI, and Cartesia voice cloning; planning HIPAA-safe logging workflows.
- Focused on backend orchestration of conversation state, audio queueing, and real-time streaming to ensure consistent user experience.
- Project highlights skills in LLM integration, accessibility-driven design, and cloud-based voice architecture.